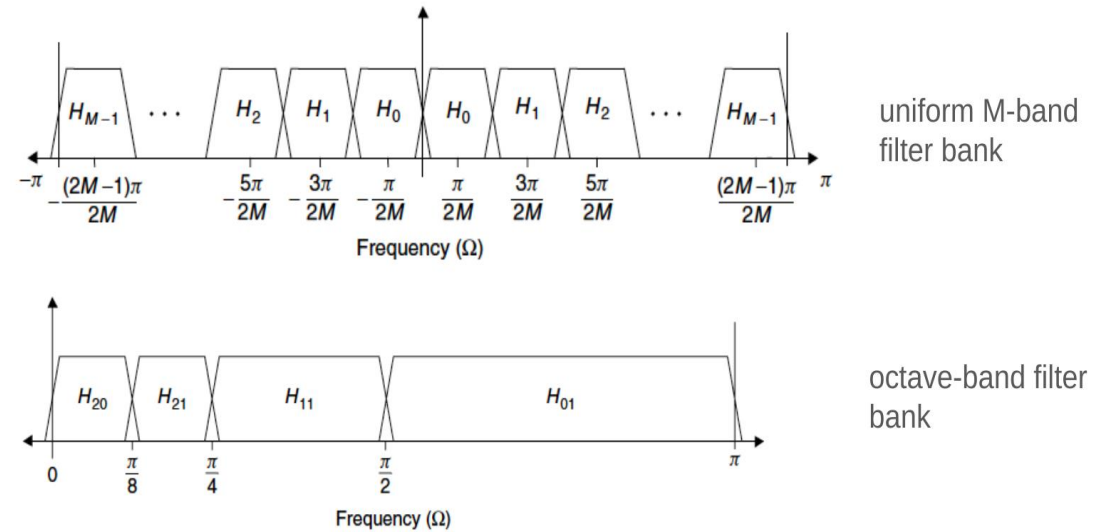


3. Time-frequency representations

Filter banks and transforms

From filters to compact time-frequency representations for audio coding.

Magnitude frequency responses of filter banks



Learning goals

- Understand FIR/IIR filters through their frequency response.
- Explain downsampling, upsampling and anti-aliasing/interpolation filters.
- Describe analysis-synthesis filter banks and critical sampling.
- Relate DFT, DCT, STFT and MDCT to audio coding.
- Discuss time-frequency tradeoffs and transient handling.

Finite Impulse Response filters

$$y[n] = \sum_{k=0}^M b_k x[n-k] = \sum_{k=0}^M h[k] x[n-k]$$

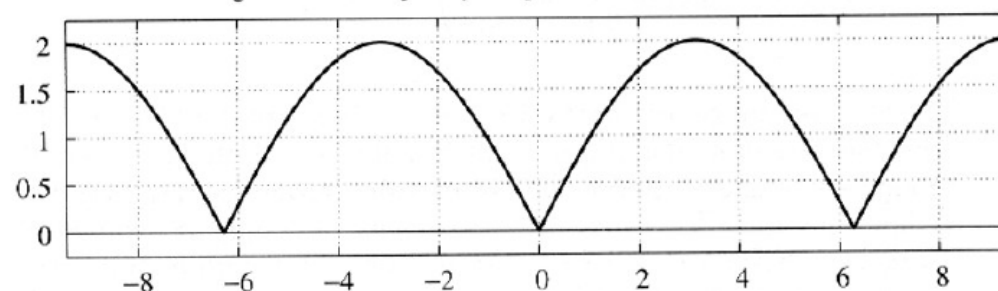
$$H(\hat{\omega}) = \sum_{k=0}^M b_k e^{-j\hat{\omega}k} = \sum_{k=0}^M h[k] e^{-j\hat{\omega}k}$$

$$y[n] = x[n] - x[n-1] \quad H(\hat{\omega}) = 1 - e^{-j\hat{\omega}} = 1 - \cos \hat{\omega} + j \sin \hat{\omega}$$

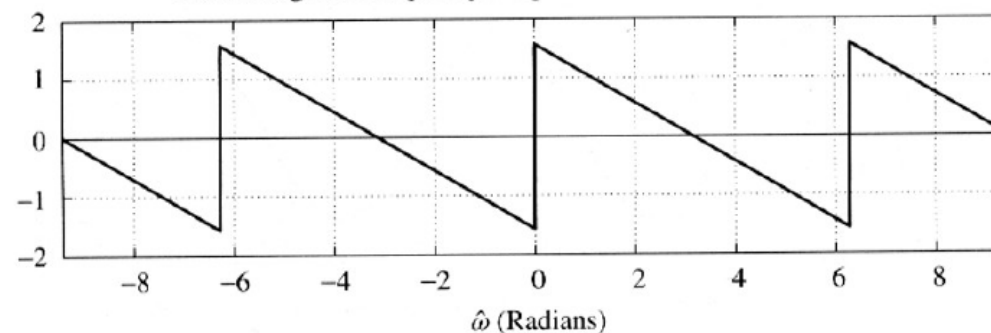
$$|H(\hat{\omega})| = \left[(1 - \cos \hat{\omega})^2 + \sin^2 \hat{\omega} \right]^{1/2} = [2(1 - \cos \hat{\omega})]^{1/2} = 2 \left| \sin \frac{\hat{\omega}}{2} \right|$$

$$\angle H(\hat{\omega}) = \arctan \left(\frac{\sin \hat{\omega}}{1 - \cos \hat{\omega}} \right)$$

Magnitude of Frequency Response of First-Difference Filter



Phase-Angle of Frequency Response of First-Difference Filter



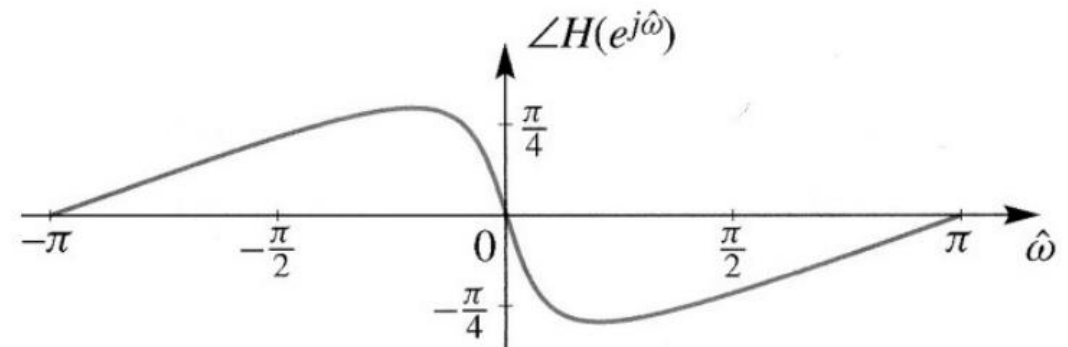
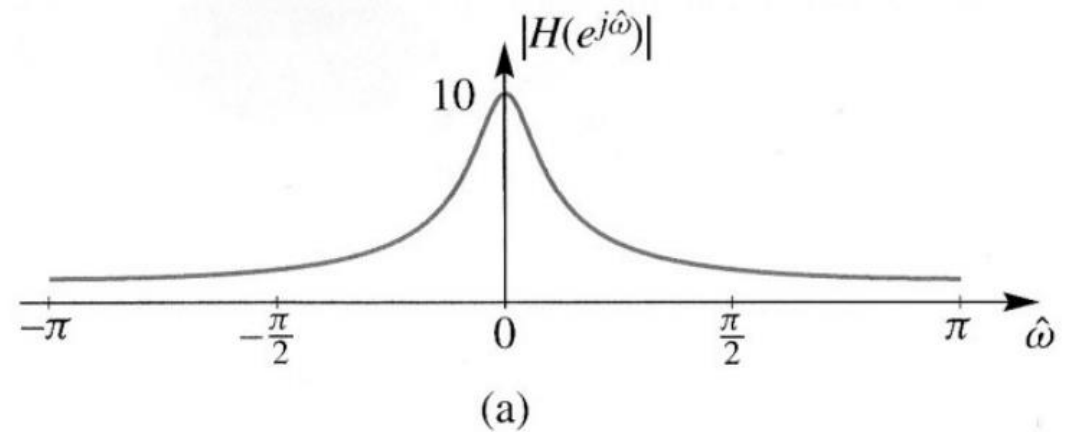
Infinite Impulse Response filters

$$H(z) = \frac{1}{1 - 0.8z^{-1}}$$

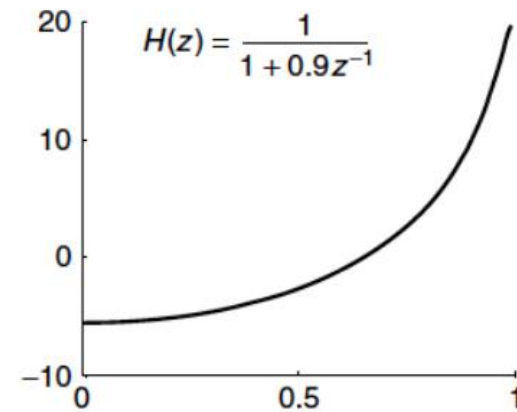
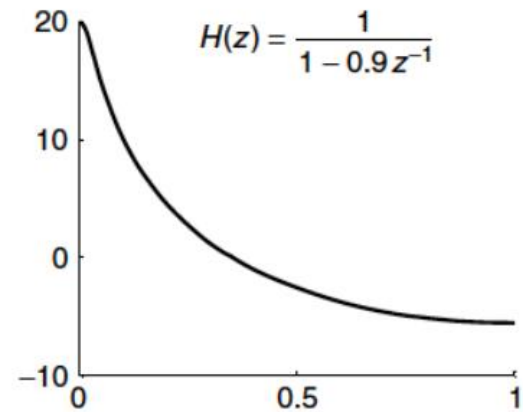
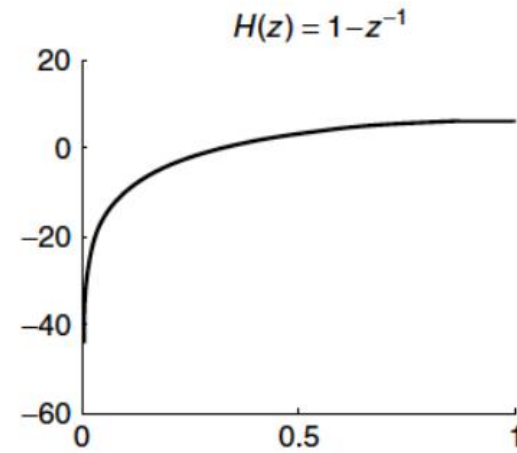
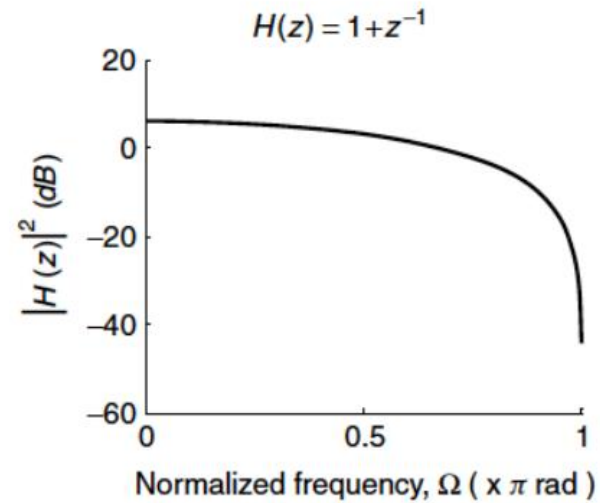
$$y[n] = \sum_{l=1}^N a_l y[n-l] + \sum_{k=0}^M b_k x[n-k]$$

$$y[n] = h[n] * x[n] \Leftrightarrow Y(z) = H(z)X(z)$$

$$H(\hat{\omega}) = H(e^{j\hat{\omega}}) = H(z) \Big|_{z=e^{j\hat{\omega}}}$$



FIR and IIR examples



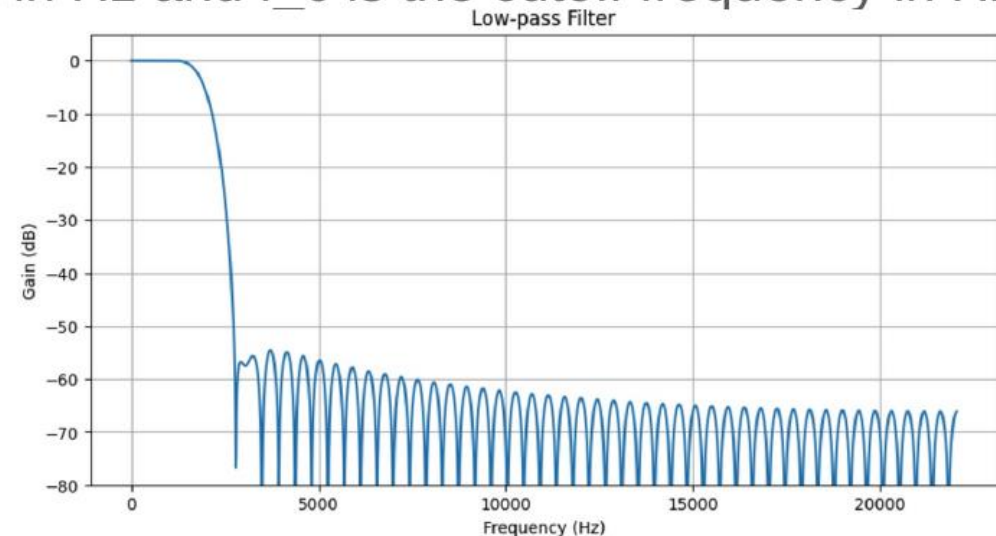
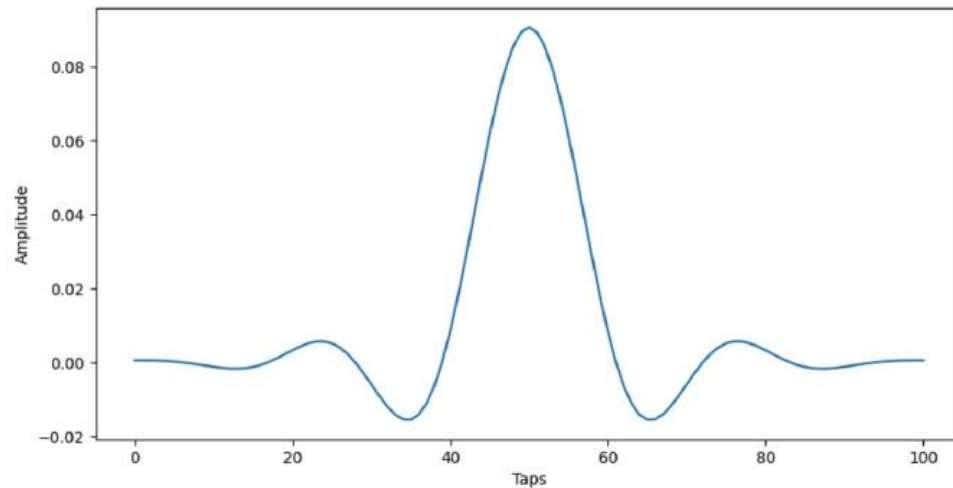
Filter design: window method (`firwin`)

Windowing a theoretically ideal impulse response $h(n)$ by some suitable chosen window function $w(n)$, yielding $h_w(n) = w(n) \cdot h(n)$, $n \in \mathbb{Z}$.

Ex: the ideal impulse response of a lowpass filter in the sinc function

$$h(n) = B \cdot \text{sinc}(Bn) \triangleq B \frac{\sin(\pi Bn)}{\pi Bn}, \quad n \in \mathbb{Z},$$

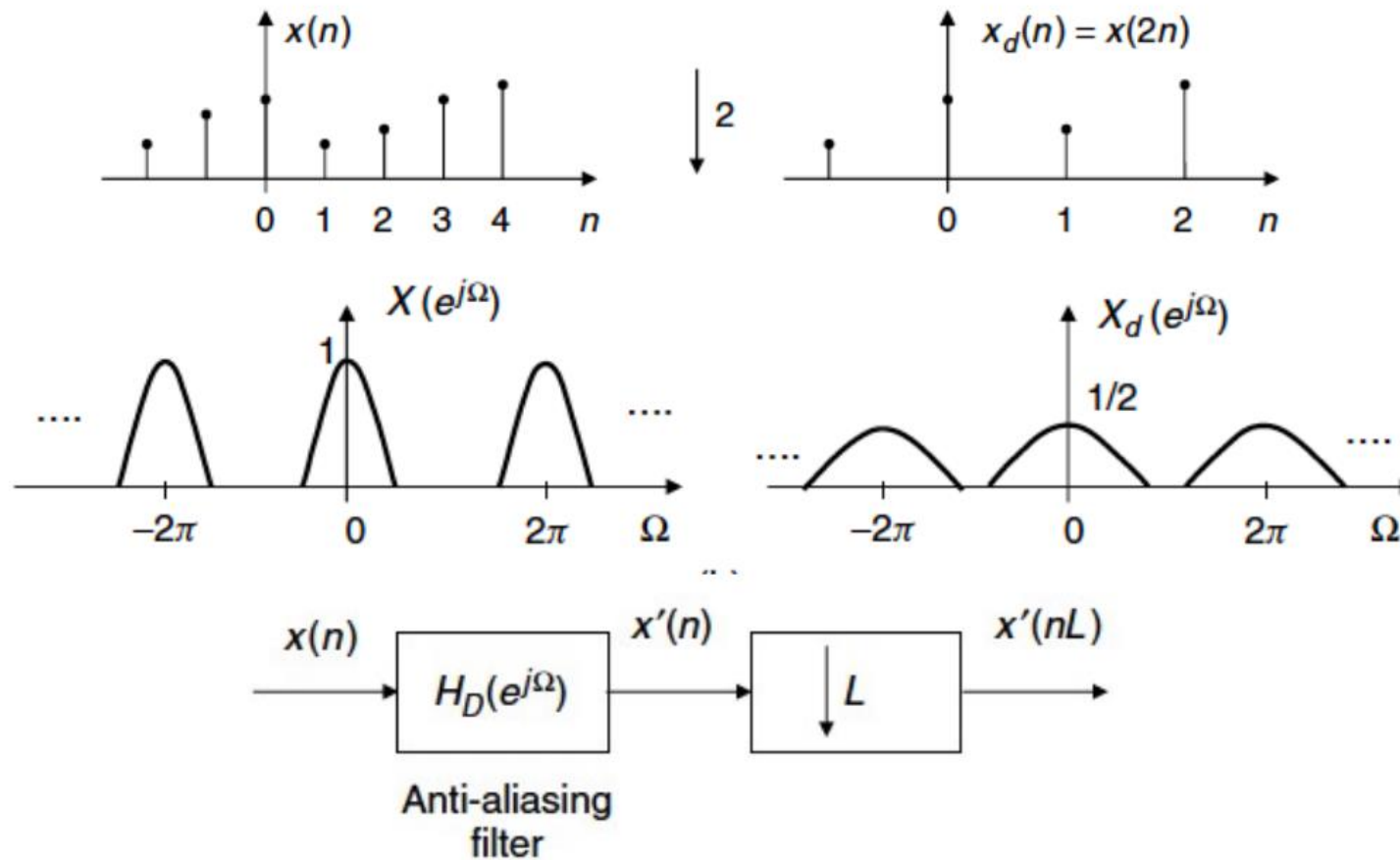
where $B = 2f_c$ is the normalized bandwidth in Hz and f_c is the cutoff frequency in Hz.



filter length = 101, $f_c = 2000$ Hz

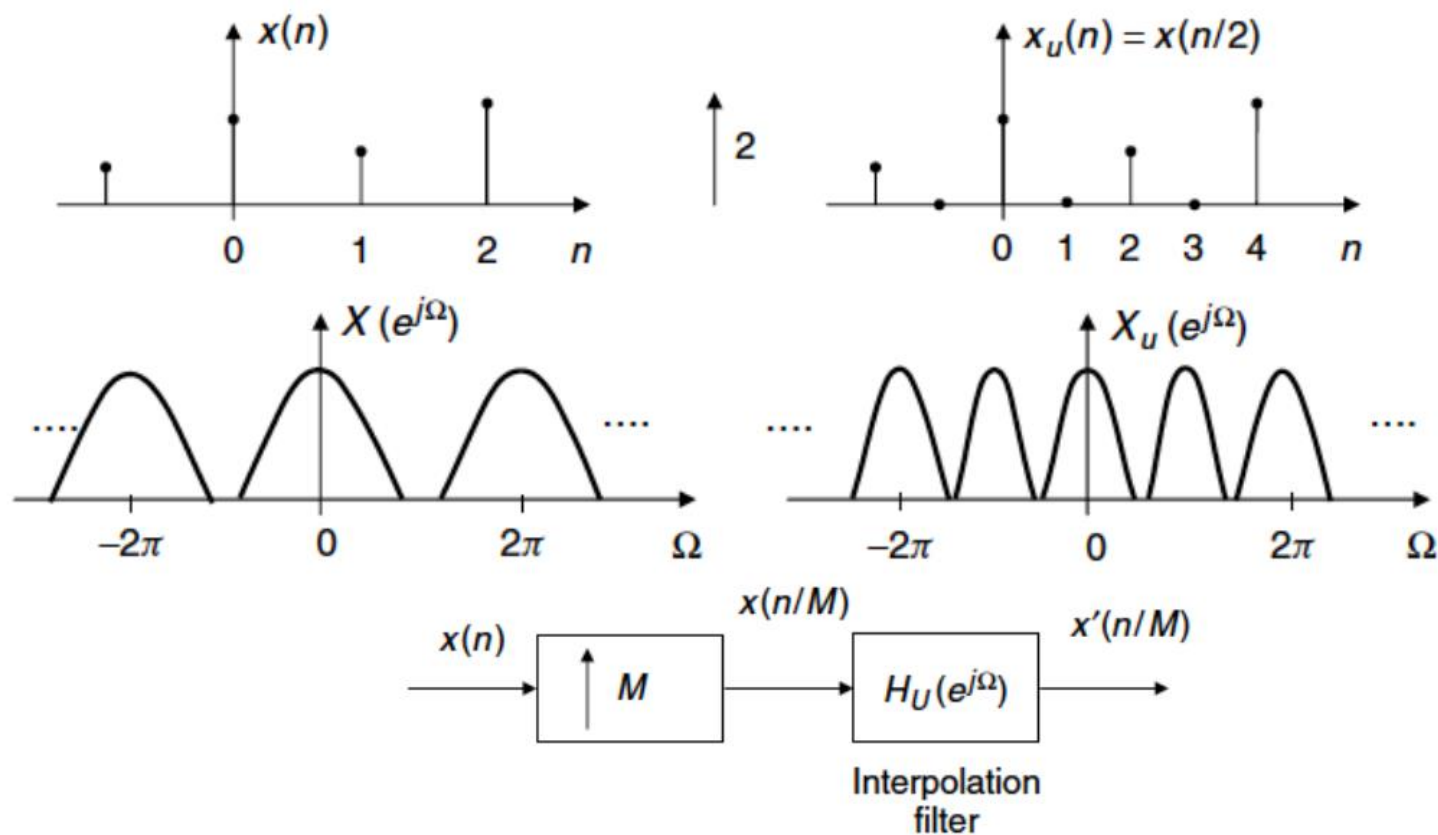
Downsampling

$$x_d(n) = x(nL)$$

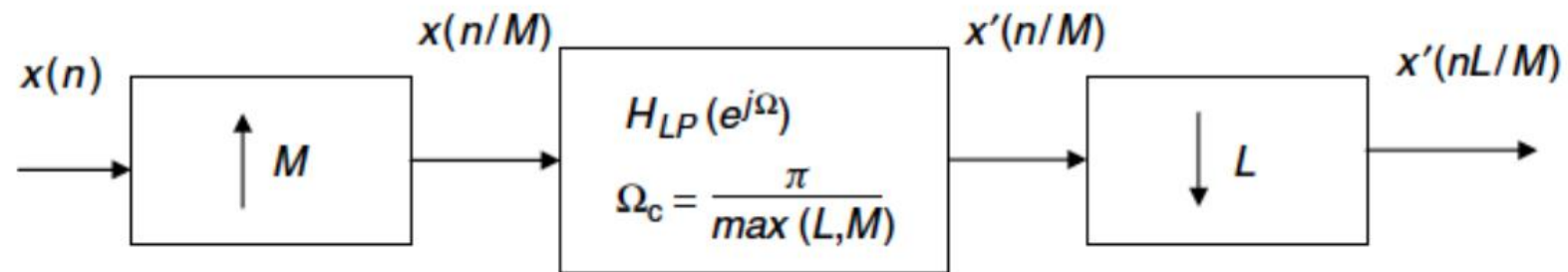


Upsampling

$$x_u(n) = \sum_{m=-\infty}^{\infty} x(m)\delta(n - mM) = \begin{cases} x(n/M), & n = 0, \pm M, \pm 2M \dots \\ 0, & \text{else} \end{cases}$$



Sampling rate changes by non-integer factor



Filter banks

An input signal $s(n)$ is processed at the encoder by a parallel bank of $(L - 1)$ -th order FIR bandpass filters, $H_k(z)$. The bandpass analysis outputs,

$$v_k(n) = h_k(n) * s(n) = \sum_{m=0}^{L-1} s(n - m)h_k(m), \quad k = 0, 1, \dots, M - 1$$

are decimated by a factor of M , yielding the subband sequences

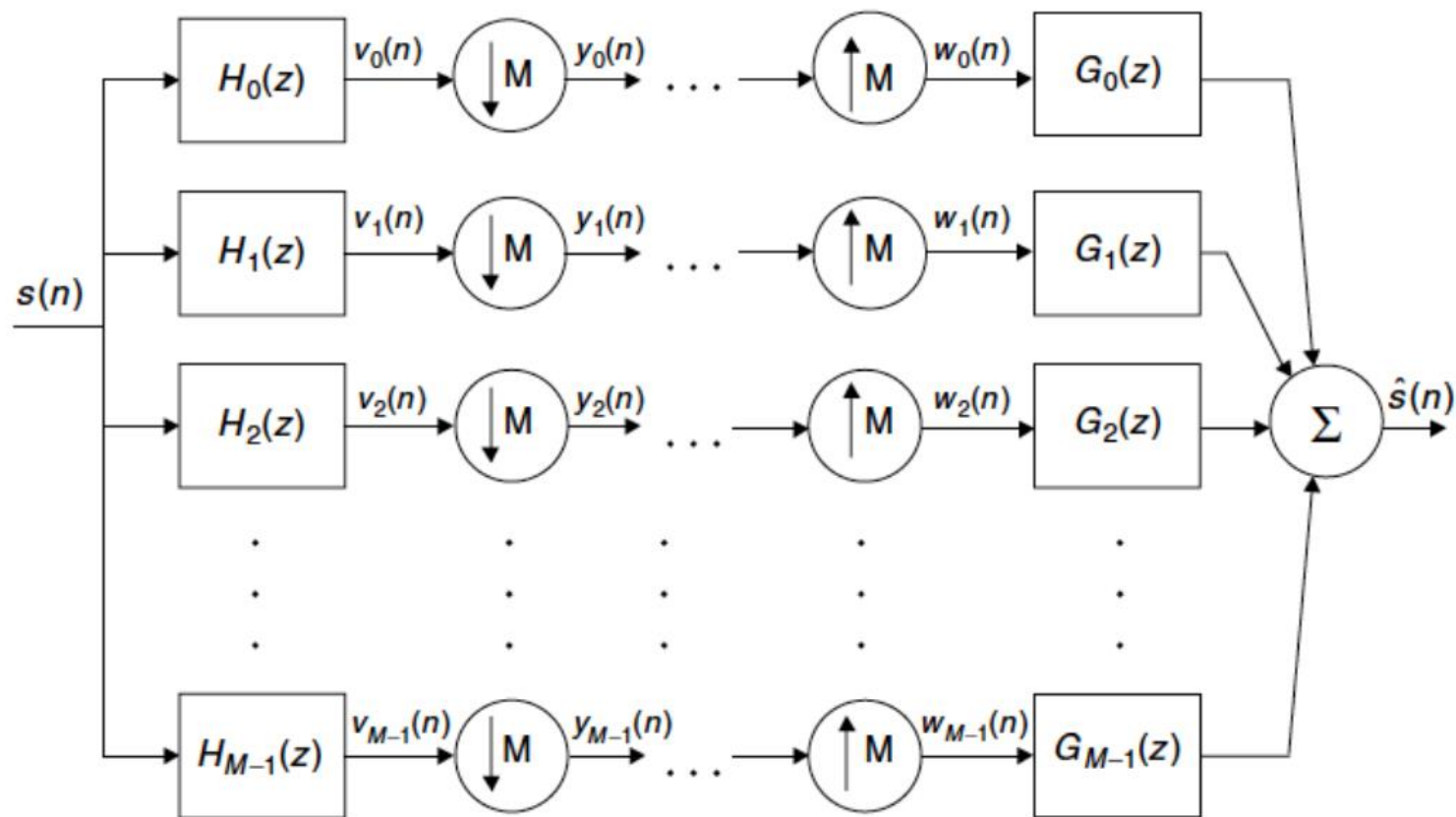
$$y_k(n) = v_k(Mn) = \sum_{m=0}^{L-1} s(nM - m)h_k(m), \quad k = 0, 1, \dots, M - 1$$

which comprise a critically sampled or maximally decimated signal representation, i.e., the number of subband samples is equal to the number of input samples.

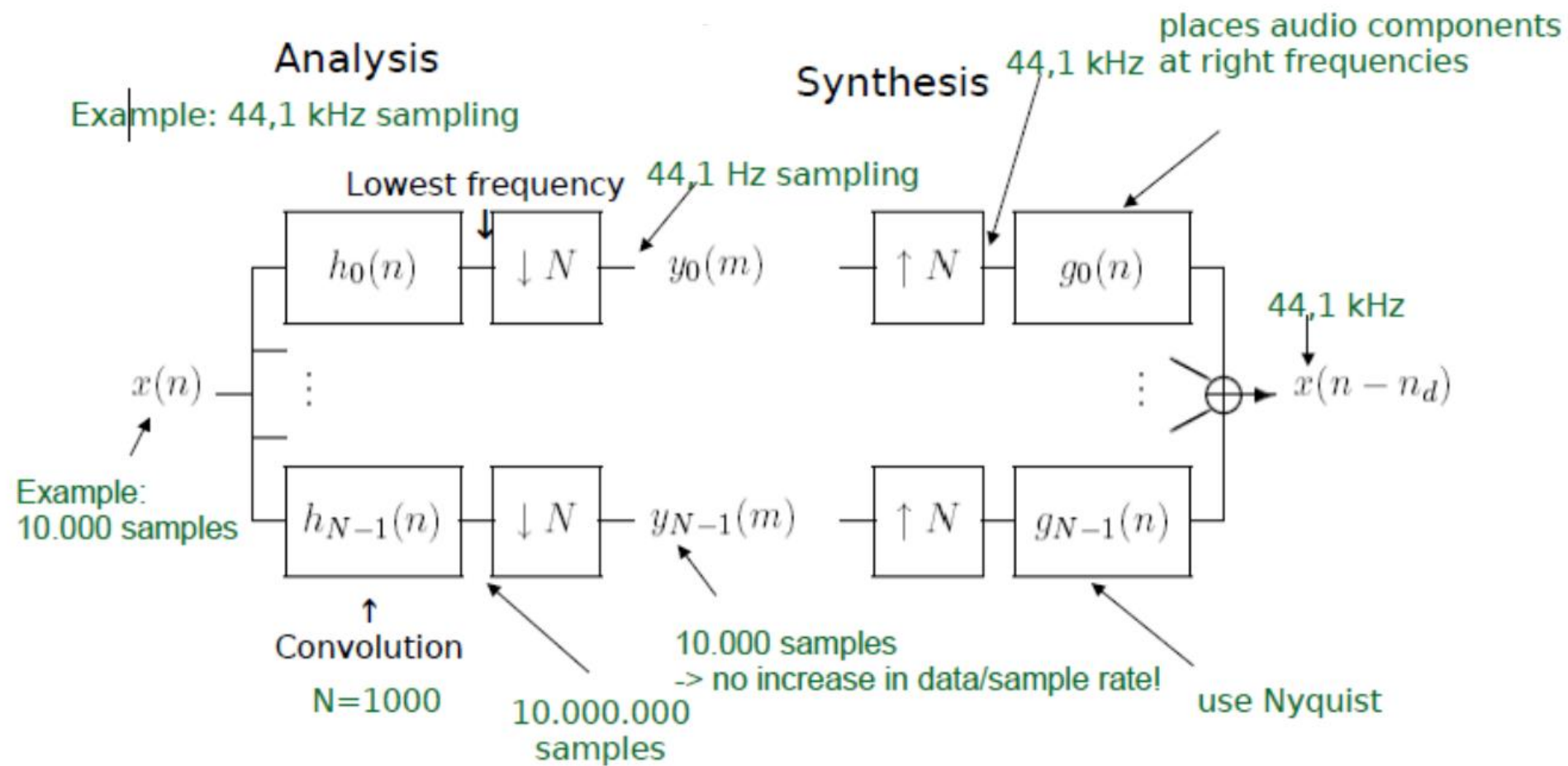
- Because it is impossible to achieve perfect “brickwall” magnitude responses with finite-order bandpass filters, there is unavoidable aliasing between the decimated subband sequences.
- Quantization and coding are performed on the subband sequences $y_k(n)$.
- The quantized subband samples are eventually received by the decoder, where they are upsampled by M .
- To eliminate the imaging distortions introduced by the upsampling the samples are processed by a parallel bank of synthesis filters, $G_k(z)$, and the filter outputs are combined to form the overall output.

$$\hat{s}(n) = \frac{1}{M} \sum_{m=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} \sum_{k=0}^{M-1} s(m) h_k(lM - m) g_k(l - Mn)$$

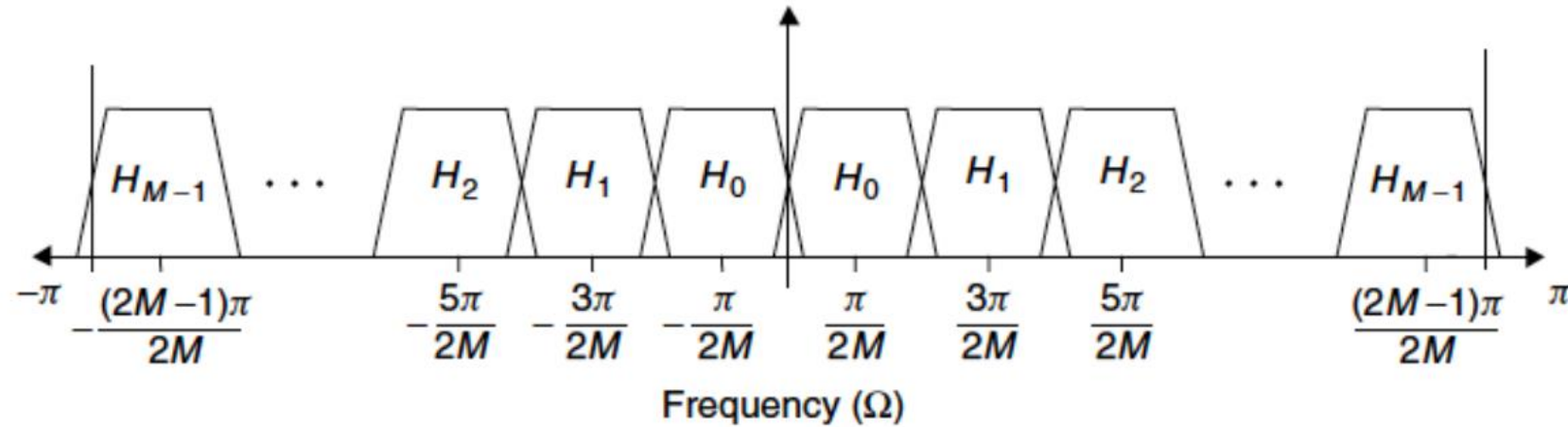
$$\hat{S}(\Omega) = \frac{1}{M} \sum_{k=0}^{M-1} \sum_{l=0}^{M-1} S\left(\Omega + \frac{2\pi l}{M}\right) H_k\left(\Omega + \frac{2\pi l}{M}\right) G_k(\Omega)$$



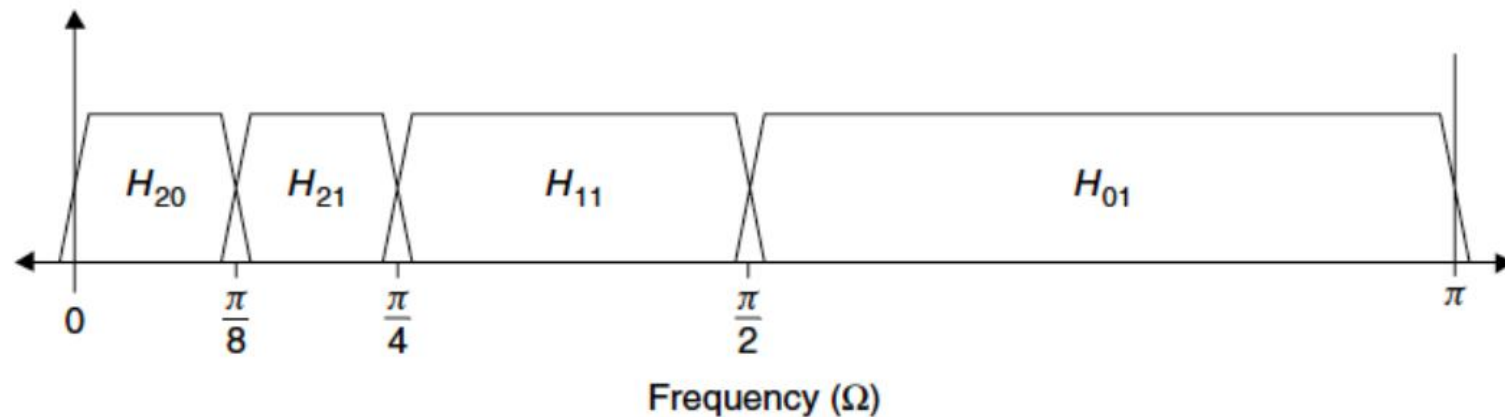
Uniform M-band maximally decimated analysis-synthesis filter bank.



Magnitude frequency responses of filter banks



uniform M-band
filter bank



octave-band filter
bank

Ideal filters

- Ideal filters are not realizable.
- In the time domain they would mean a convolution of our signal with a Sinc function.
- Sinc function is infinitely long and not causal, meaning it causes infinite delay.
- We cannot simply use a DFT or FFT to obtain an ideal filter in the frequency domain either. Because the DFT also represents a filter bank, but a special type.
- Its equivalent filters are far from perfect filters (hence we cannot make ideal filters with it), not good enough for our purposes (audio coding and the ear).

Cosine modulated filter banks

- Cosine modulation of a lowpass prototype filter can realize parallel M-channel filter banks with nearly perfect reconstruction (called “Pseudo QMF”).
- Analysis and synthesis filters satisfy the mirror image condition

$$g_k(n) = h_k(L - 1 - n)$$

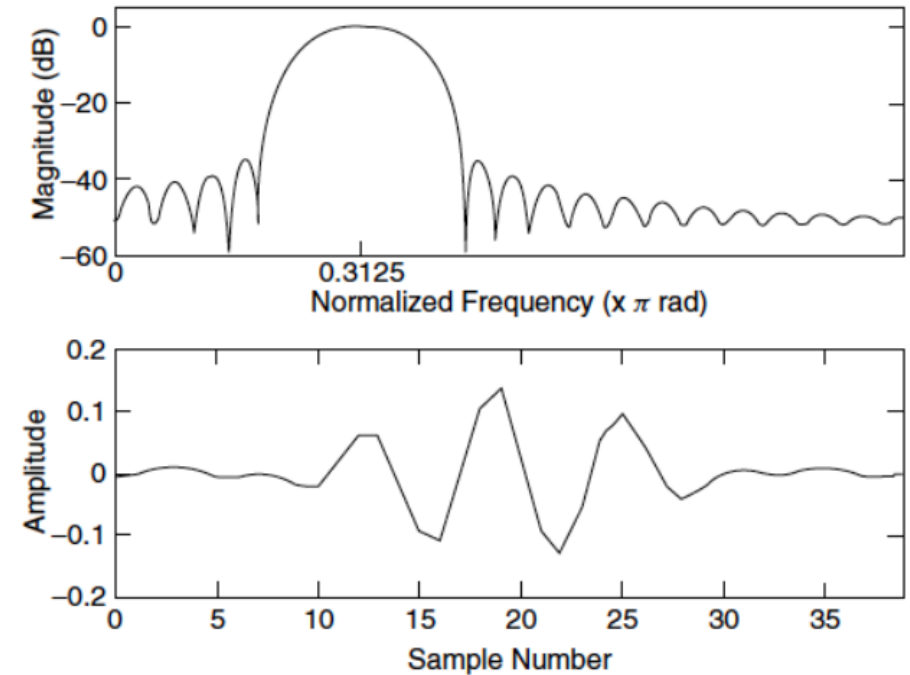
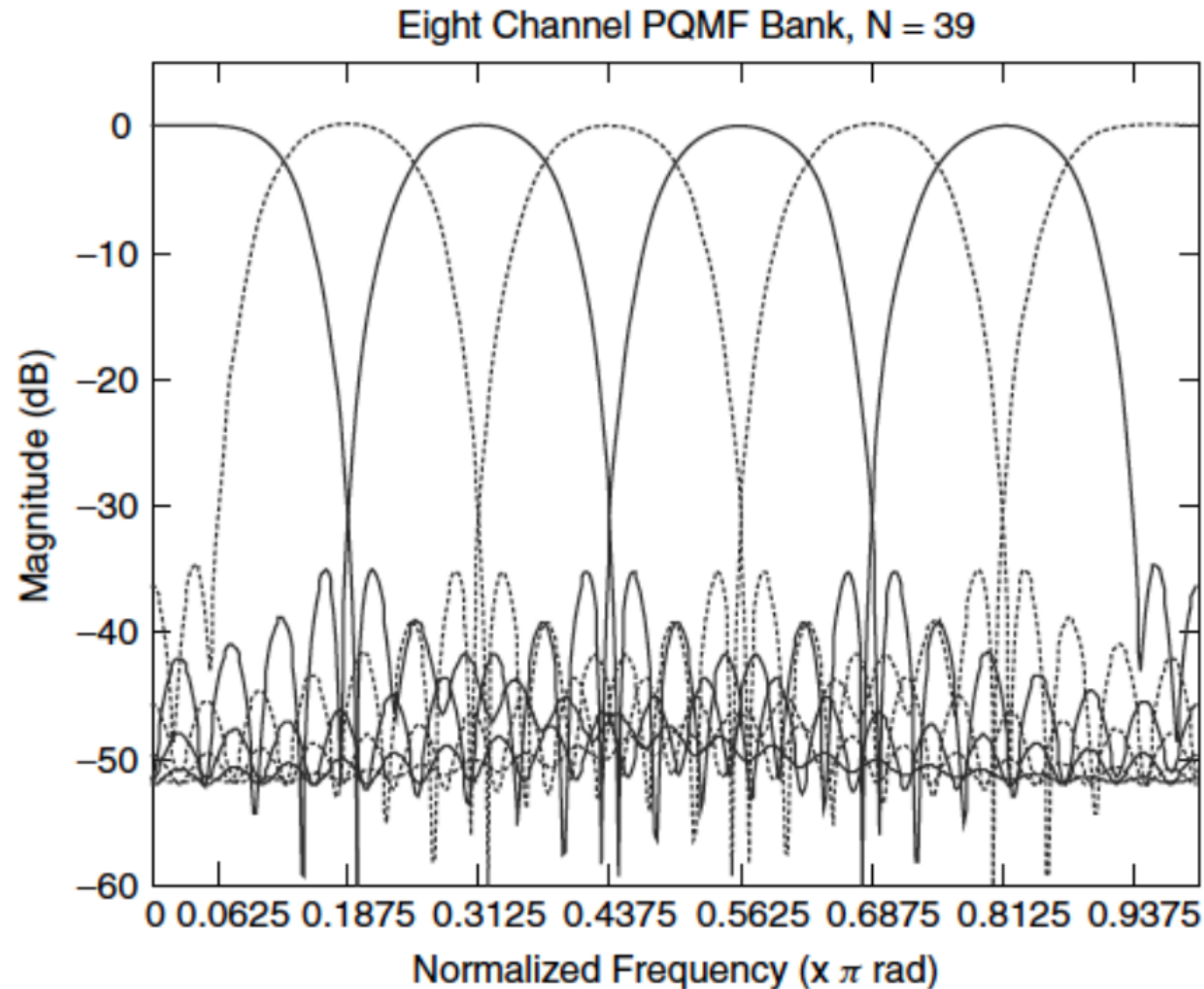
$$h_k(n) = 2w(n) \cos \left[\frac{\pi}{M}(k + 0.5) \left(n - \frac{(L - 1)}{2} \right) + \Omega_k \right]$$

$$g_k(n) = 2w(n) \cos \left[\frac{\pi}{M}(k + 0.5) \left(n - \frac{(L - 1)}{2} \right) - \Omega_k \right]$$

$$\text{where, } \Omega_k = (-1)^k \frac{\pi}{4}$$

and the sequence $w(n)$ corresponds to the L-sample “window”, a real-coefficient linear phase FIR prototype lowpass filter, with normalized cutoff frequency $\pi/2M$.

Pseudo Quadrature Mirror Filters (PQMF)



Magnitude frequency response and impulse response for channel 3.

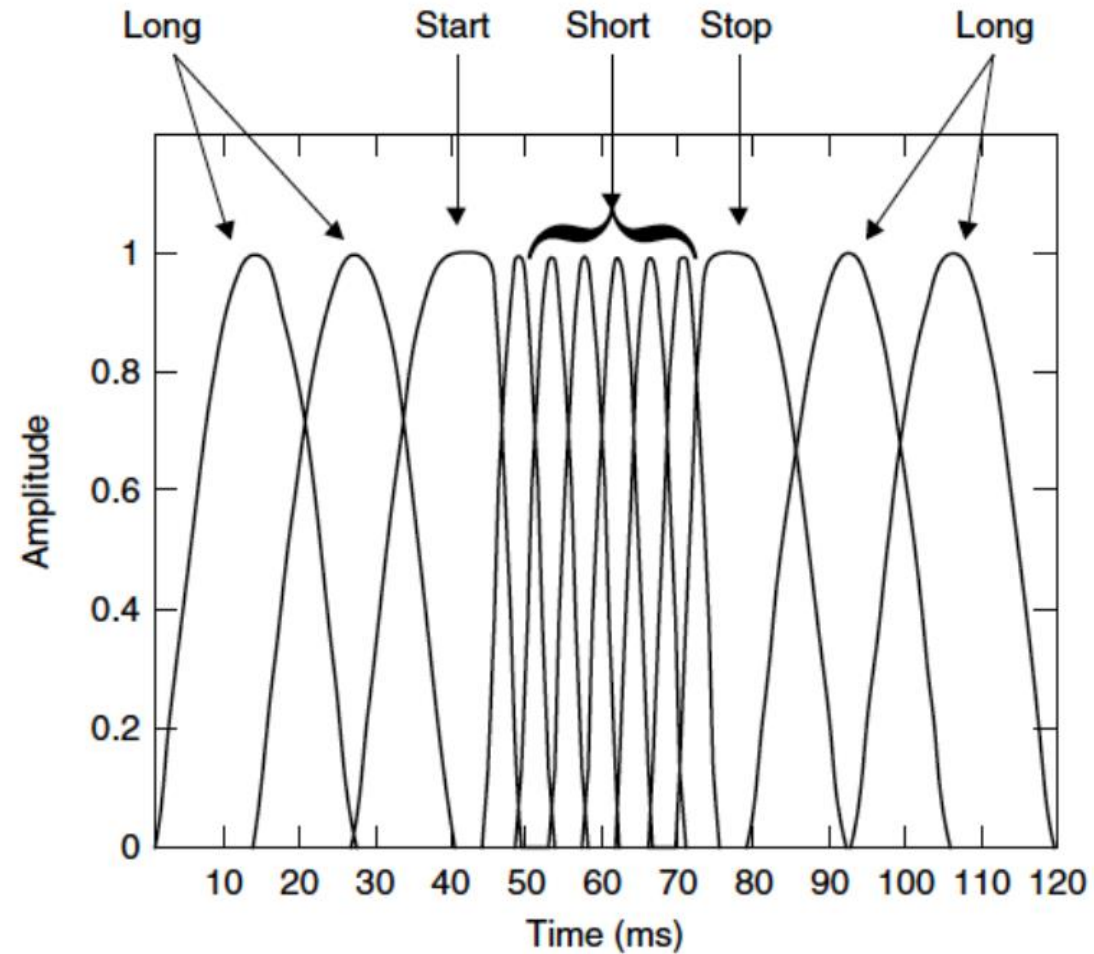
PQMF properties

- Constrained design: single FIR prototype filter
- Uniform, linear phase channel responses
- Overall linear phase, hence constant group delay
- Low complexity, i.e. one filter plus modulation
- Amenable to fast block algorithms
- Critical sampling

Implementation

- ***Analysis Filter Bank***: The input audio signal is passed through a bank of analysis filters, which splits the signal into multiple subbands.
- ***Downsampling***: Each subband is then downsampled to reduce the data rate.
- ***Encoding/Decoding (Optional)***: Quantization or other encoding techniques can be applied to each subband for compression.
- ***Upsampling***: The downsampled subbands are upsampled back to their original rate.
- ***Synthesis Filter Bank***: The upsampled subbands are passed through a synthesis filter bank to reconstruct the audio signal.

Hybrid, switched filter banks



Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j 2\pi kn/N} \quad k=0, \dots, N-1$$

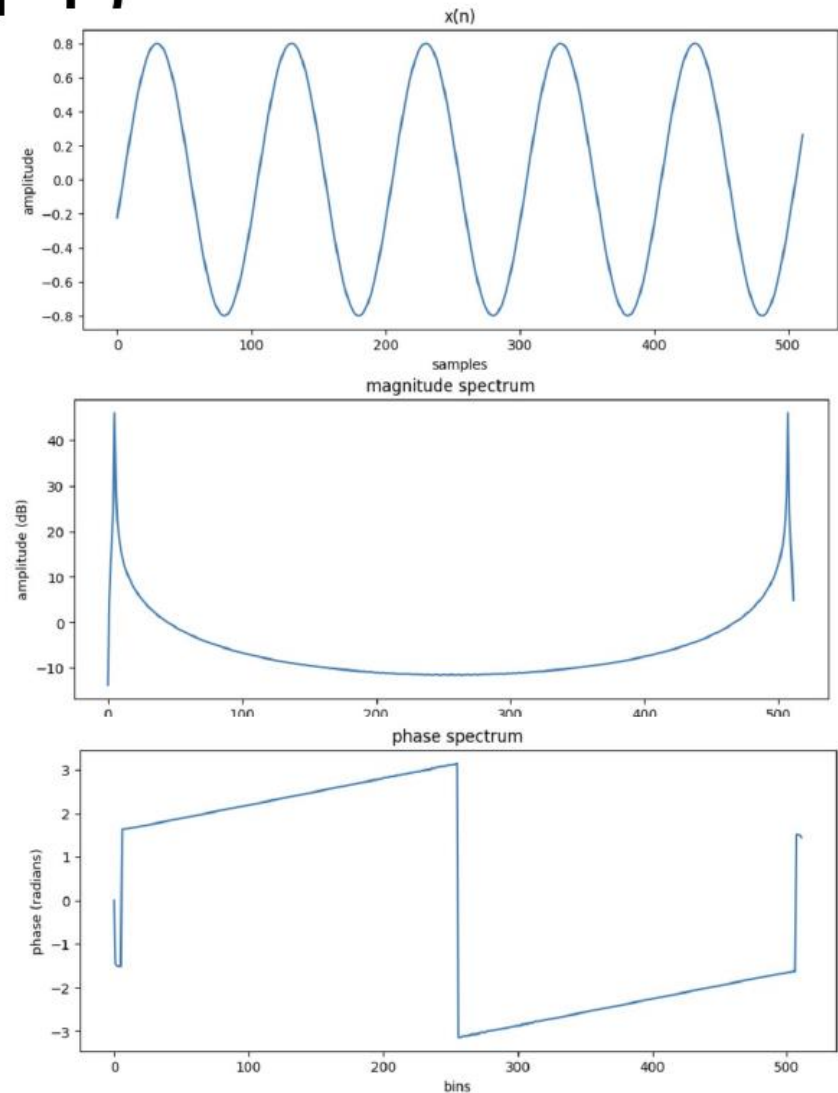
n : discrete time index (normalized time, $T=1$)

k : discrete frequency index

$\omega_k = 2\pi k/N$: frequency in radians/second

$f_k = f_s k/N$: frequency in Hz (f_s : sampling rate)

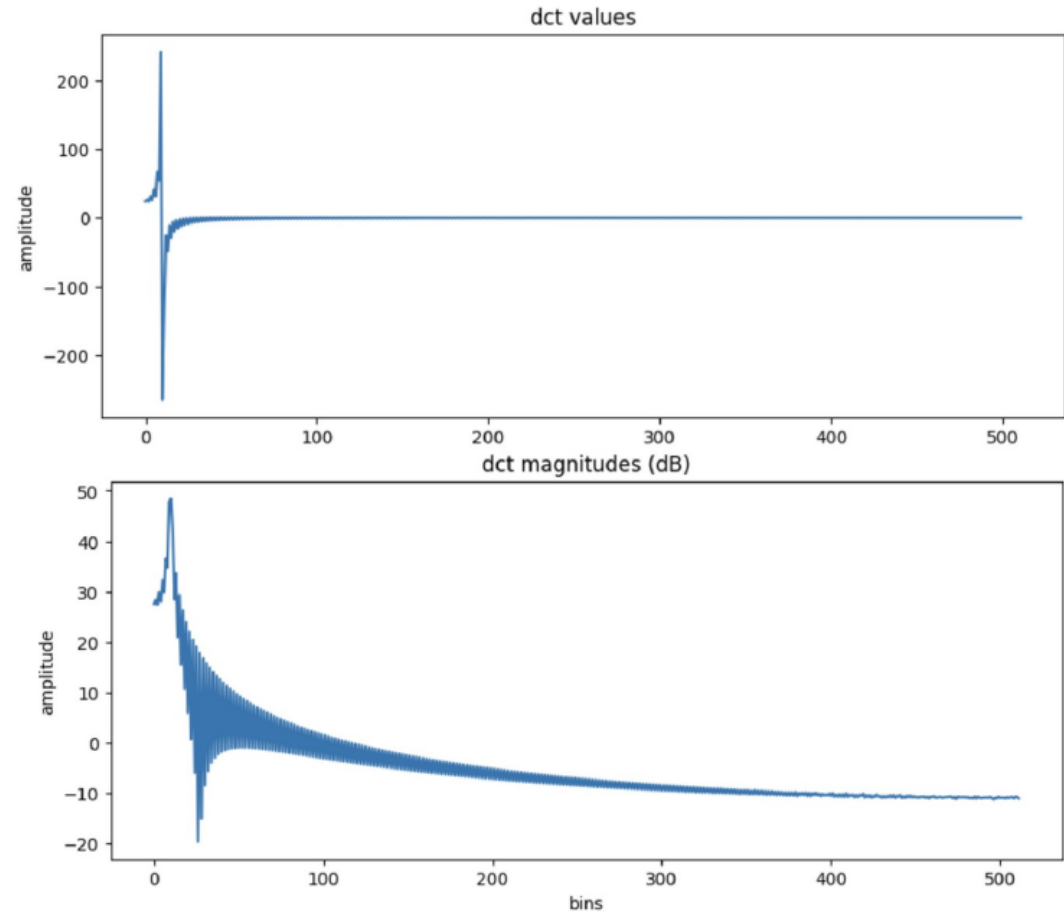
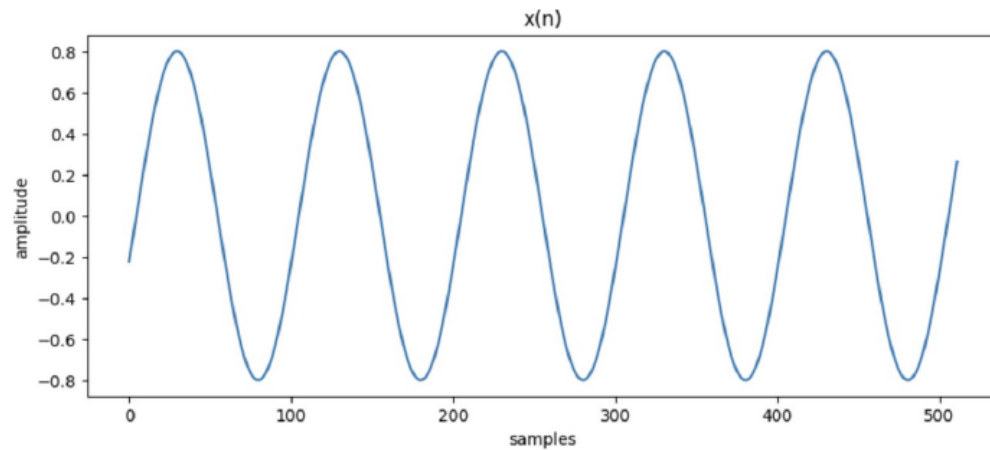
$$x[n] = \sum_{k=0}^{N-1} X[k] e^{j 2\pi kn/N} \quad n=0, 1, \dots, N-1$$



Discrete Cosine Transform (DCT)

$$y_k(m) = \sqrt{\frac{2}{N}} \sum_{n=0}^{N-1} x(mN + n) \cdot \cos\left(\frac{\pi}{N}(n+0.5)(k+0.5)\right)$$

DCT Type 4



DFT vs DCT

- The DFT is the basic frequency-domain transformation, conceptually easy to understand, and suitable for most kind of spectral analysis and spectral transformations. However, DFT requires overlapping between windows, thus it doubles the information that need to be stored: not very good for compression.
- DCT returns a real-valued spectrum (only the positive frequencies). It has twice as many frequency values for the same number of time-domain samples.

Short-time Fourier Transform (STFT)

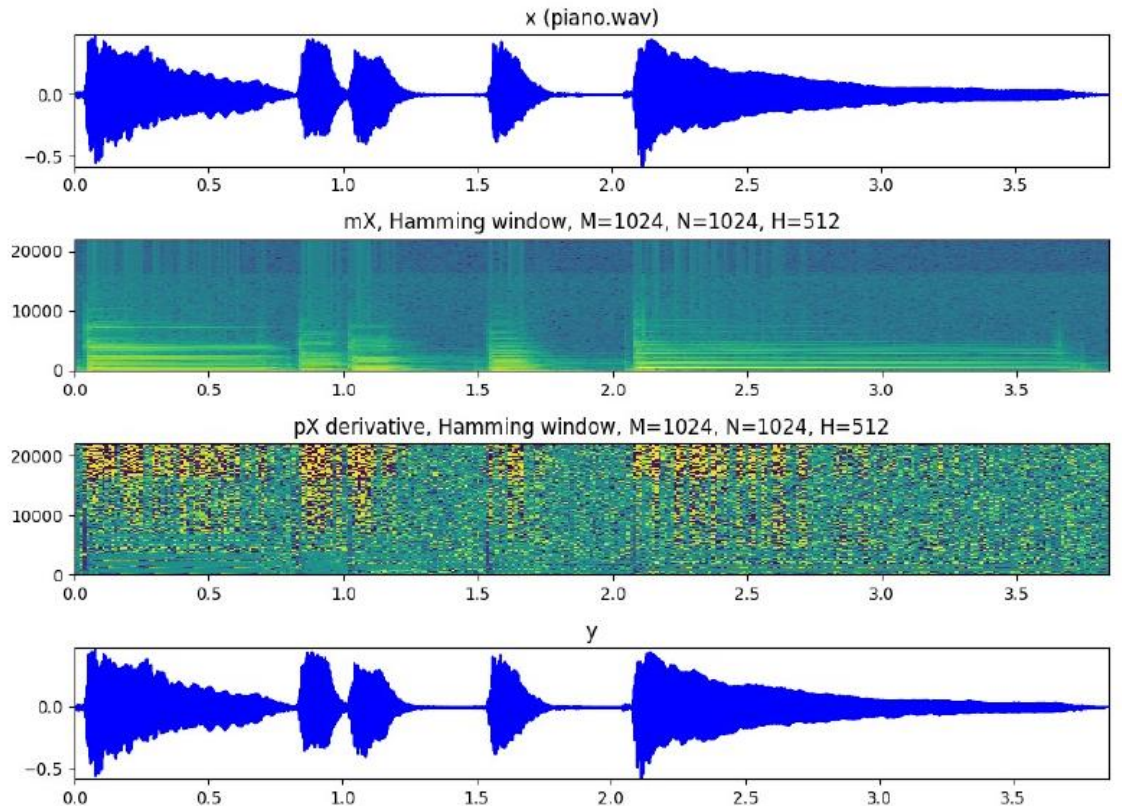
$$X_l[k] = \sum_{n=-N/2}^{N/2-1} w[n] x[n+lH] e^{-j2\pi kn/N} \quad l=0,1,\dots,$$

w : analysis window

l : frame number

H : hop-size

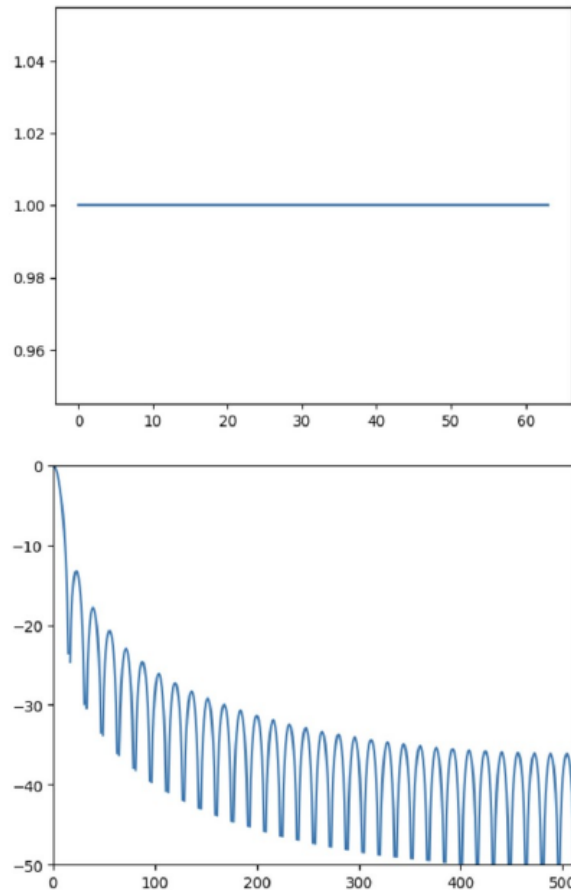
$$y[n] = \sum_{l=0}^{L-1} \text{Shift}_{lH,n} \left[\frac{1}{N} \sum_{k=-N/2}^{N/2-1} X_l[k] e^{j2\pi kn/N} \right]$$



Windows

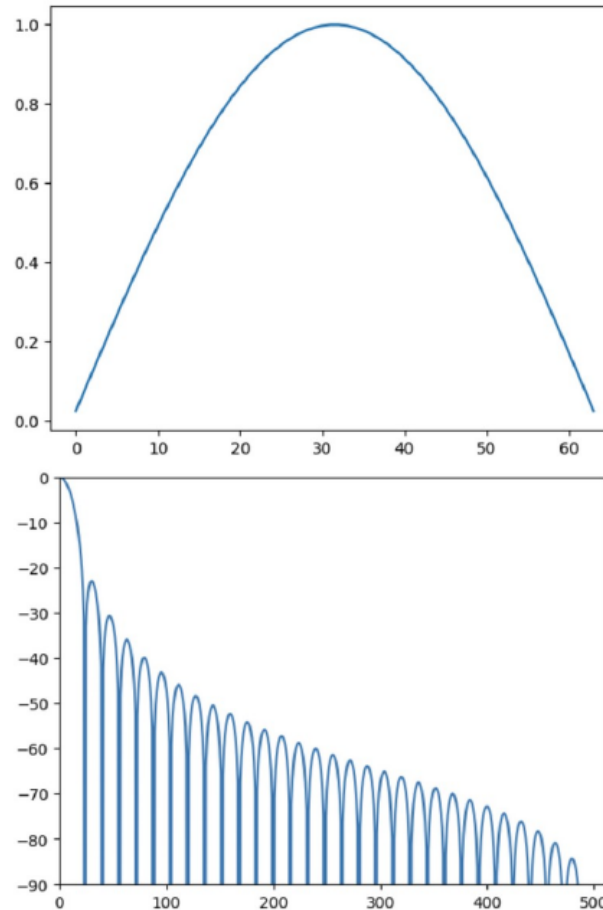
Rectangular window:

$$w[n]=1$$



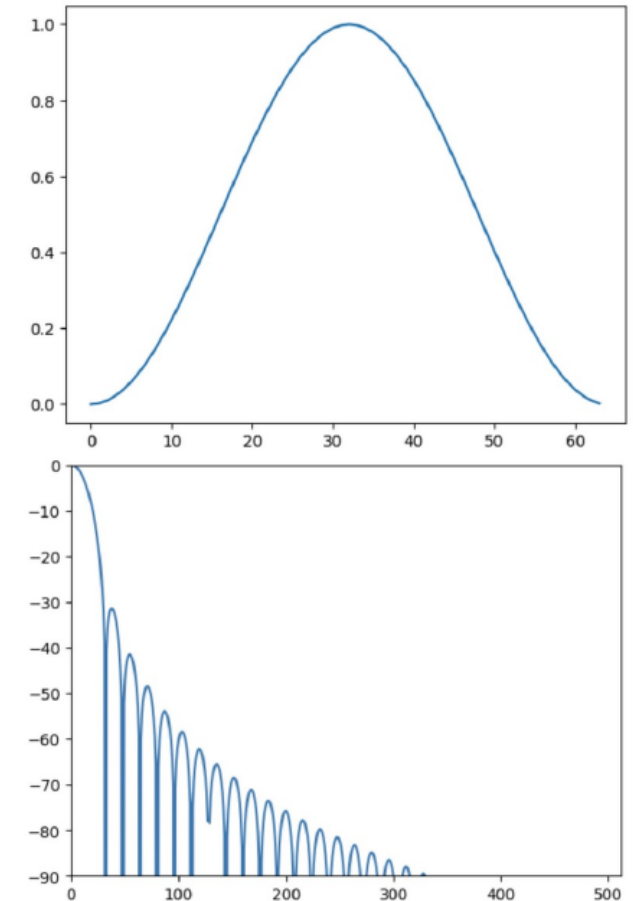
Sine window:

$$w(n) = \sin \left[\left(n + \frac{1}{2} \right) \frac{\pi}{2M} \right]$$



Hanning window:

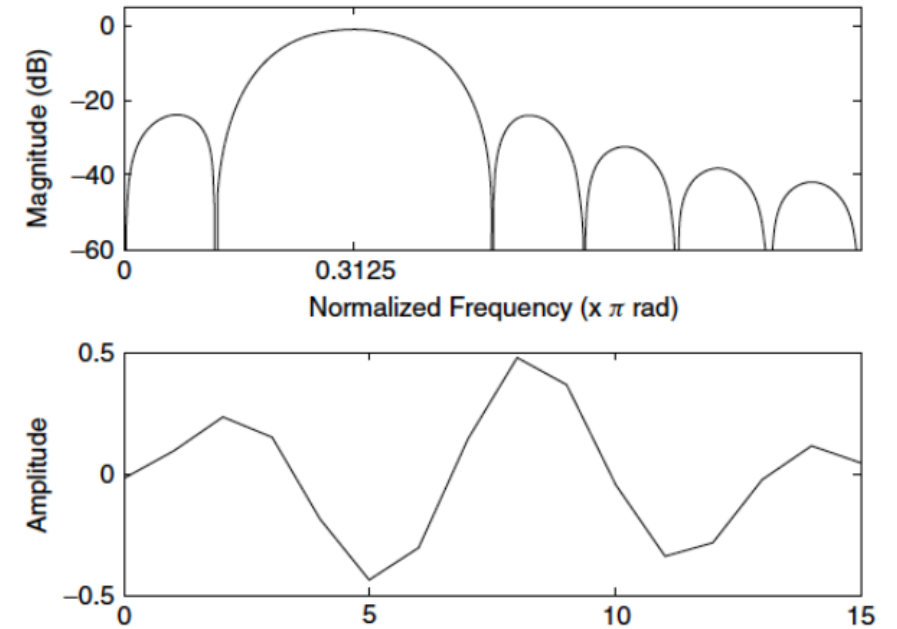
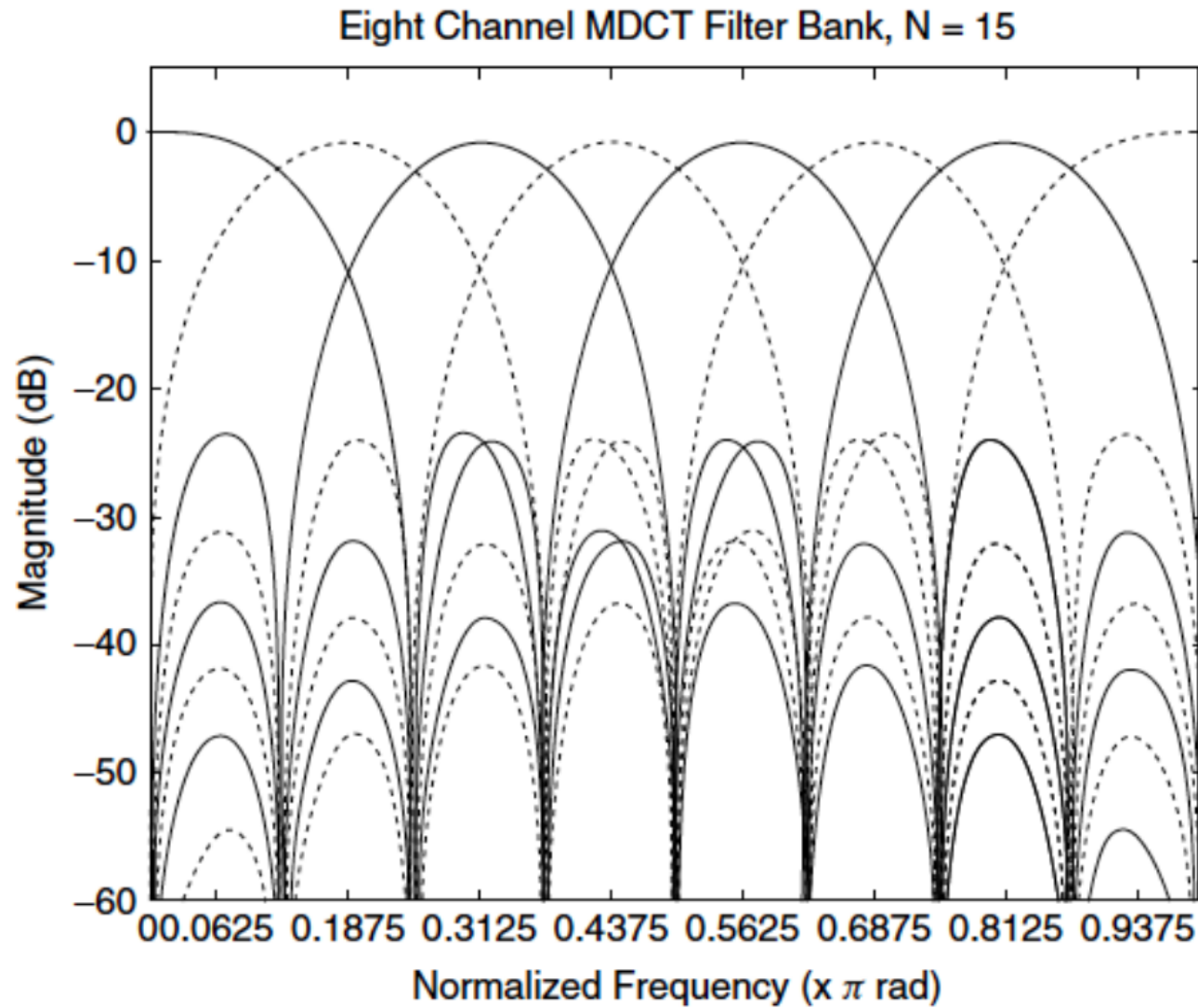
$$w[n] = .5 + .5 \cos(2\pi n/M)$$



Modified Discrete Cosine Transform (MDCT)

- Solves the data rate problem inherent in DFT coders. No window overlap.
- It is a class of Time Domain Alias Cancellation (TDAC) transforms.
- It does not invert like the DFT to recover the input signal. It includes time-domain aliasing which is cancelled in the time domain. Thus it allows for perfect reconstruction.
- It can be implemented with the FFT.

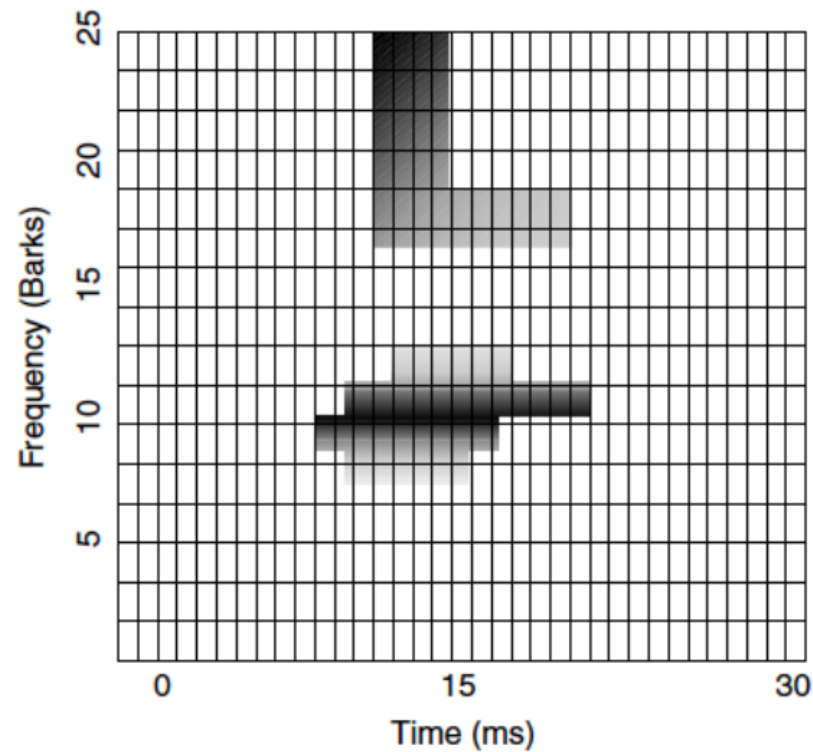
$$h_k(n) = h(n) \cdot \sqrt{\frac{2}{N}} \cdot \cos\left(\frac{\pi}{N} \left(k + \frac{1}{2}\right) \left(n + \frac{N}{2} + \frac{1}{2}\right)\right)$$



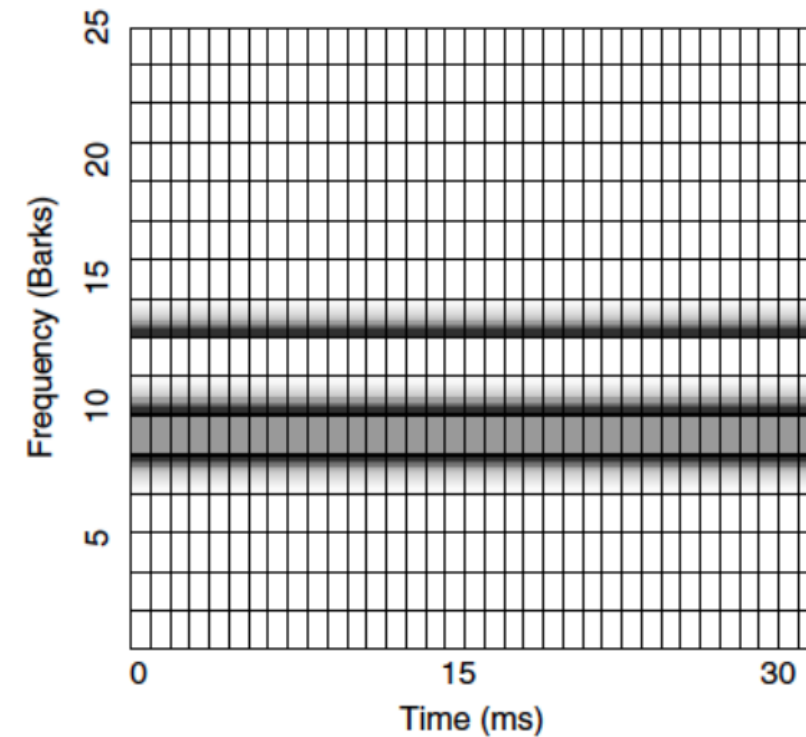
Magnitude frequency response and impulse response for channel 3.

Time-frequency resolution issues

Difficulty of selection a single filter bank to satisfy the diverse time and frequency resolutions requirements associated with different classes of audio.



castanets

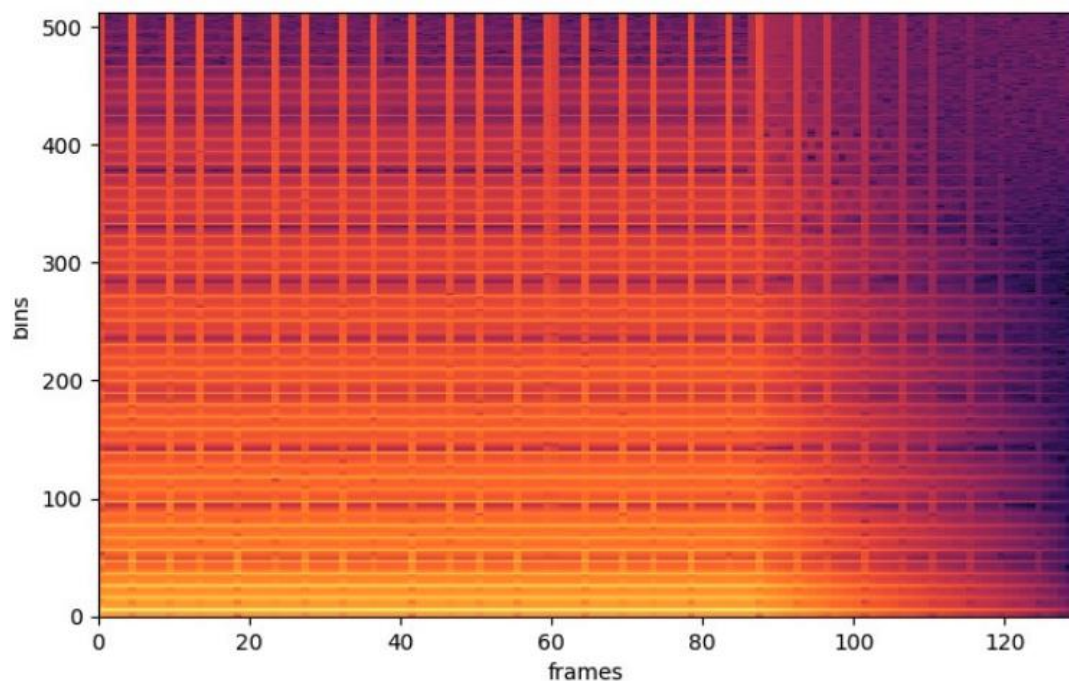


piccolo

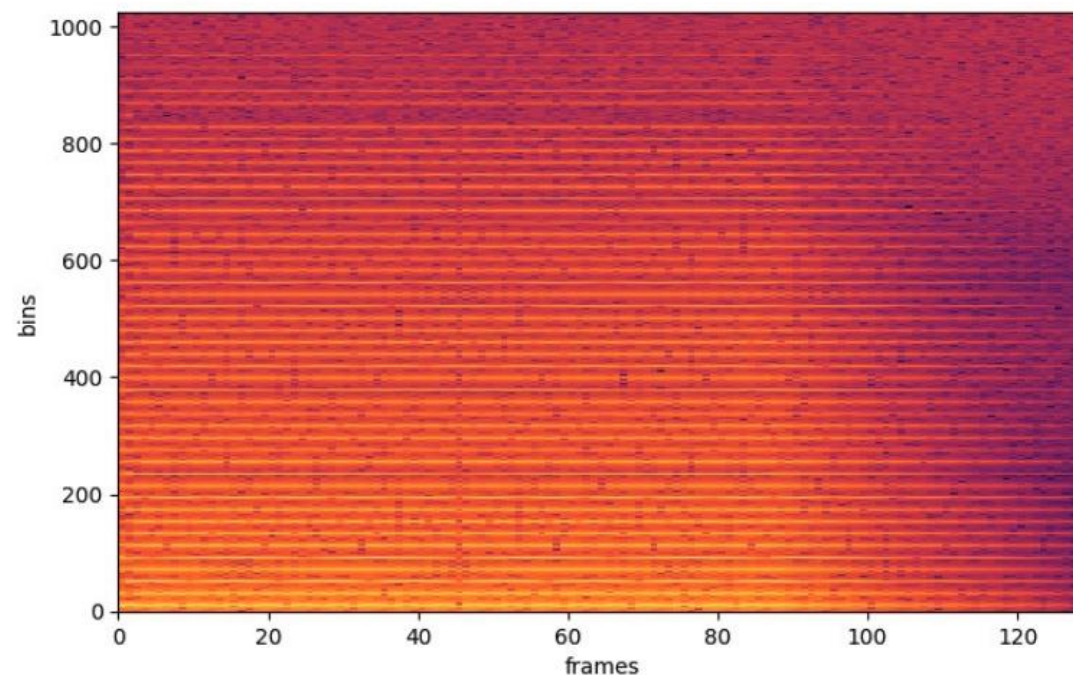
STFT vs MCDT

- STFT built on top of DFT and MCDT built on top of DCT Type 4
- MCDT has not window overlap, it is critically sampled

STFT, no window overlap



MCDT



Key takeaways

- Audio coders need compact representations that expose perceptually meaningful frequency bands.
- Downsampling and upsampling require careful filtering to control aliasing and imaging.
- Filter banks split audio into subbands, but perfect reconstruction is a design constraint.
- MDCT is central in perceptual coding because it supports critical sampling and time-domain alias cancellation.
- Window length controls the resolution tradeoff: transients need time precision; steady tones need frequency precision.

References and notebooks

Core reading

Andreas Spanias, Ted Painter, Venkatraman Atti (2007). Audio Signal Processing and Coding. Chapter 6: Time-frequency analysis: filter banks and transforms.

Course notebooks

- `examples/FIR-design.ipynb`
- `examples/filter-bank-basic.ipynb`
- `examples/filter-bank-critically-sampled.ipynb`
- `examples/dft-dct.ipynb`
- `examples/dct-mcdt.ipynb`
- `examples/windows.ipynb`
- `examples/stft-mdct.ipynb`

github.com/xserra/audio-coding-materials